OptionLib: A Modular Python Library for Pricing European, Asian, and American Stock Options via Monte Carlo Simulation

ISyE 6644 — Simulation and Modeling for Engineering and Science
Georgia Institute of Technology — Summer 2026

Project Topic 27: Stock Option Pricing — "A Library of Programs to Price Different Types of Stock Options"

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Abstract

We develop OptionLib, a modular Python library that prices several classes of equity options and quantifies the statistical quality of every estimate it produces. Under the Black–Scholes–Merton (BSM) risk-neutral framework, we implement (i) closed-form pricing and Greeks for European options as an analytical benchmark, (ii) a geometric-Brownian-motion (GBM) Monte Carlo (MC) engine for European, arithmetic-Asian, and American options, and (iii) a suite of variance-reduction and sensitivity tools. Across ten strike/type combinations, naïve MC prices agree with BSM within Monte Carlo error, and 95% confidence intervals cover the true price in a validation study (empirical coverage 0.950–0.953 against a nominal 0.95). A geometric-average control variate reduces the variance of the arithmetic-Asian estimator by roughly 1,300×, and common random numbers (CRN) reduce finite-difference Delta variance by roughly 1,100×. Longstaff–Schwartz least-squares MC prices an American put to within about 0.7% of a 3,000-step binomial benchmark and correctly recovers a positive early-exercise premium. All experiments use a fixed replication budget with reported seeds. The result is a validated, reproducible option-pricing toolkit and a case study in applying core simulation concepts—CIs, variance reduction, CRN, and benchmarking against ground truth.

1 Background and Description of the Problem

1.1 The option-pricing problem

A financial option is a contract that grants its holder the right—but not the obligation—to buy (a call) or sell (a put) an underlying asset at a fixed strike price K. The central problem of quantitative finance is to assign a fair present value to that right. Under the risk-neutral valuation principle, the price of an option is the expected value of its discounted payoff under a risk-neutral probability measure $V = E_Q[\, e^{-rT} \cdot payoff(S)\,]$, where r is the risk-free rate and T the maturity. For a European call the payoff is $max(S_T - K, 0)$; for a European put it is $max(K - S_T, 0)$. This expectation is the quantity our library estimates, either in closed form when one exists or by Monte Carlo simulation when it does not.

Options differ by their payoff structure and exercise style. European options may be exercised only at maturity. Asian options pay on the average price of the underlying over the option's life, which reduces sensitivity to end-of-period price manipulation but destroys the simple lognormal structure that makes European options analytically tractable. American options may be exercised at any time up to maturity, introducing an optimal-stopping problem that has no closed-form solution in general. Our library targets all three styles because together they exercise the full range of simulation techniques taught in this course: direct Monte Carlo, path simulation, variance reduction, common random numbers, and regression-based dynamic programming.

1.2 Why simulation

The BSM model gives a closed-form price for European options (Black and Scholes, 1973; Merton, 1973), which we use as ground truth. But the moment a payoff becomes path-dependent (Asian) or the exercise policy becomes a decision problem (American), closed forms either vanish or require restrictive assumptions. Monte Carlo simulation, introduced to option pricing by Boyle (1977), is the general-purpose remedy: it estimates $E_Q[\cdot]$ by simulating many risk-neutral price paths, averaging discounted payoffs, and forming confidence intervals. Its cost is statistical noise that decays only as $O(n\text{^}\{-1/2\})$; the practical art, and much of the content of this report, is reducing that noise through variance reduction (Boyle, Broadie, and Glasserman, 1997) and handling early exercise through least-squares Monte Carlo (Longstaff and Schwartz, 2001).

1.3 Past efforts and literature

The problem has a rich literature. Black and Scholes (1973) and Merton (1973) derived the parabolic PDE and closed-form European price. Cox, Ross, and Rubinstein (1979) introduced the binomial lattice, which converges to BSM and, crucially, extends to American options by comparing continuation and exercise values at every node—we use a fine (up to 3,000-step) lattice as our American benchmark. Boyle (1977) pioneered Monte Carlo pricing; Boyle, Broadie, and Glasserman (1997) surveyed variance-reduction methods including antithetic and control variates. Kemna and Vorst (1990) gave the geometric-average Asian price in closed form, which we exploit as a control variate for the (analytically intractable) arithmetic-average Asian option. Longstaff and

Schwartz (2001) proposed least-squares Monte Carlo (LSM) for American options, regressing realized continuation values on basis functions of the state to approximate the optimal exercise boundary. Glasserman (2004) is the standard reference tying these threads together. Our contribution is not new theory but a validated, reproducible implementation that quantifies the statistical behavior of each estimator against known ground truth—precisely the simulation-rigor emphasis of this course.

1.4 Organization of the report

Section 2 states the mathematical model and input parameters. Section 3 describes the library architecture and how to run it. Section 4—the analytical core—reports five experiments: (4.1) validation of the MC engine against BSM with a CI-coverage study; (4.2) a variance-reduction study for European and Asian options; (4.3) American pricing via LSM benchmarked against the binomial lattice; (4.4) Greeks by analytical formulas versus finite-difference MC; and (4.5) the effect of common random numbers on sensitivity estimation. Section 5 concludes and proposes future work. Code provenance and an AI-usage disclosure appear in the appendices.

2 Mathematical Model and Inputs

2.1 Risk-neutral dynamics

We assume the underlying follows geometric Brownian motion under the risk-neutral measure Q: $dS_t = r \cdot S_t\, dt + \sigma \cdot S_t\, dW_t$, where W_t is a standard Brownian motion, r is the constant risk-free rate, and σ the constant volatility. The exact solution is lognormal:

$$S_t = S_0 \cdot exp[\,(r - \frac{1}{2}\sigma^2) \cdot t + \sigma \cdot \sqrt{t} \cdot Z\,],\ \ Z \sim N(0,1).$$

For path-dependent products we discretize [0,T] into m equal steps of size Δt = T/m and evolve $S_{t+\Delta t} = S_t \cdot exp[(r - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t} \cdot Z]$. Because GBM has an exact lognormal transition, this scheme introduces no time-discretization bias in the marginal law of S_t; the only discretization effect is that the discrete average used by an Asian payoff differs from a continuous average, which we account for in the matching closed-form control variate.

2.2 Closed-form benchmarks

For a European call/put the BSM price is

$$C = S_0 \cdot N(d_1) - K \cdot e^{-rT} \cdot N(d_2), \; P = K \cdot e^{-rT} \cdot N(-d_2) - S_0 \cdot N(-d_1),$$

$$d_1 = [ln(S_0/K) + (r + ½\sigma^2)T] / (\sigma\sqrt{T}), \; d_2 = d_1 - \sigma\sqrt{T},$$

where N(·) is the standard normal CDF. The analytical Greeks (Delta, Gamma, Vega, Theta, Rho) follow by differentiation and provide ground truth for Section 4.4. For the geometric-average Asian option we use the Kemna–Vorst (1990) closed form, in which the geometric average of a lognormal process is itself lognormal with an adjusted drift and volatility; this exact price anchors the control-variate estimator in Section 4.2.

2.3 Inputs and parameterization

Because this is a programming-oriented topic, our inputs are generated from known distributions with known parameters so that estimator accuracy can be measured against ground truth—the rubric's preferred design for non-application projects. The single stochastic input is the standard-normal shock Z driving GBM, generated with NumPy's PCG64 generator. Table 1 lists the base contract and market parameters, chosen as a canonical textbook configuration (at-the-money, one-year, moderate volatility) so results are comparable with published values; strikes are then swept from deep in- to deep out-of-the-money.

Parameter	**Value**
Spot S_0	100
Strike K	{80,90,100,110,120}
Risk-free rate r	0.05 (5%)
Volatility σ	0.20 (20%)
Maturity T	1.0 year
Dividend yield q	0.0
Time steps m (path)	50 (Asian/LSM)
Replications n	10^5 per condition
RNG	NumPy PCG64, seeded

Table 1. Base model and simulation parameters used throughout the experiments.

3 Library Architecture and Implementation

OptionLib is organized as a small pipeline of composable modules, each with a single responsibility. This decomposition keeps the simulation core independent of any specific payoff and lets every pricer reuse the same path generator, variance-reduction hooks, and RNG discipline.

Module	Responsibility
analytics.py	Closed-form BSM prices, Greeks, and the Kemna–Vorst geometric-Asian price (benchmarks).
gbm.py	GBM path/terminal-price simulator; optional antithetic sampling; seed control.
mc_european.py	Terminal-price MC pricer for European calls/puts with CI and optional control variate.
mc_asian.py	Path-based arithmetic-Asian pricer with geometric-average control variate.
american.py	Binomial (CRR) lattice reference and Longstaff–Schwartz LSM for American options.
greeks.py	Analytical Greeks and finite-difference MC Greeks with common random numbers.
varred.py	Antithetic / control-variate estimators and variance-reduction diagnostics.

Table 2. OptionLib module architecture; each pricer reuses the shared GBM simulator and RNG.

3.1 Simulation core and reproducibility

Every stochastic routine accepts an explicit NumPy Generator so results are reproducible from a reported seed. A subtlety we document explicitly: for the common-random-numbers and control-variate experiments the same draw of Z must be reused across the compared conditions, whereas for the coverage study each macro-replication must draw fresh, independent shocks. We therefore pass the RNG object in (not a bare seed) so the caller controls exactly when streams are shared versus advanced. The MC estimator and its standard error are computed as

$$\hat{V} = (1/n)\, \Sigma\, e\hat{\ }\{-rT\} \cdot payoff_i, \;\; SE = s\,/\,\sqrt{n}, \;\; 95\%\, CI = \hat{V} \pm 1.96 \cdot SE,$$

where s is the sample standard deviation of the discounted payoffs. We use z = 1.96 throughout for 95% intervals, justified by the large replication counts ($n \geq 10^5$), which make the sample mean effectively normal by the CLT.

3.2 Representative code

The terminal-price European pricer, with optional antithetic and control-variate switches, is the template all other pricers follow:

```python
def mc_european(S0,K,r,sigma,T,n,call=True,antithetic=False,control=False,rng=None):
    if antithetic:
        Z = rng.standard_normal(n//2); Z = np.concatenate([Z,-Z])
    else:
        Z = rng.standard_normal(n)
    ST     = S0*np.exp((r-0.5*sigma**2)*T + sigma*np.sqrt(T)*Z)
    disc   = np.exp(-r*T)*np.maximum(ST-K,0) if call else np.exp(-r*T)*np.maximum(K-ST,0)
    if control:                       # CV = discounted S_T, E_Q[e^{-rT}S_T]=S0
        cv = np.exp(-r*T)*ST
        b  = np.cov(disc,cv)[0,1]/np.var(cv,ddof=1)
        disc = disc - b*(cv - S0)
    return disc.mean(), disc.std(ddof=1)/np.sqrt(len(disc))
```

The full source (all seven modules, a README, and a reproduce.py that regenerates every table and figure in this report) is provided in the accompanying zip. To run, e.g., the validation study: python reproduce.py --experiment validate.

4 Main Findings

We report five experiments. Every experiment fixes its replication budget in advance, reports confidence intervals or standard errors, and benchmarks against a known analytical or lattice ground truth. All seeds are fixed and recorded in reproduce.py.

4.1 Validating the Monte Carlo engine against Black–Scholes

We first confirm the MC engine is unbiased by pricing ten European contracts (five strikes × {call, put}) with $n = 10^5$ replications each and comparing to the exact BSM price. Table 3 reports the true price, the MC estimate, its standard error, the 95% CI, and whether the interval covers the truth. Nine of ten intervals cover the true price; the single miss (the deep-OTM K=80 put, true value 0.687) is expected: at a nominal 95% level roughly one interval in twenty should miss, and OTM puts have the noisiest relative payoff. The estimates track BSM closely across the entire moneyness range.

K	Type	BSM (true)	MC est	SE	CI low	CI high	Covers?
80	Call	24.5888	24.6844	0.0607	24.5654	24.8035	Yes

80	Put	0.6872	0.6655	0.0083	0.6493	0.6818	No
90	Call	16.6994	16.6180	0.0550	16.5102	16.7257	Yes
90	Put	2.3101	2.3386	0.0170	2.3053	2.3719	Yes
100	Call	10.4506	10.4479	0.0465	10.3567	10.5391	Yes
100	Put	5.5735	5.6001	0.0274	5.5464	5.6539	Yes
110	Call	6.0401	6.0065	0.0366	5.9347	6.0782	Yes
110	Put	10.6753	10.6458	0.0378	10.5717	10.7200	Yes
120	Call	3.2475	3.2037	0.0272	3.1504	3.2570	Yes
120	Put	17.3950	17.3550	0.0470	17.2629	17.4470	Yes

Table 3. Naïve MC vs. BSM for ten European contracts (n = 10^5). CIs are 95% (z = 1.96).

Figure 1 shows the running MC estimate for the ATM call converging to the BSM price of 10.451 as n grows, with the 95% CI band narrowing at the characteristic $O\left(n^{-\frac{1}{2}}\right)$ rate; Figure 2 shows thirty simulated GBM paths that feed the estimator.

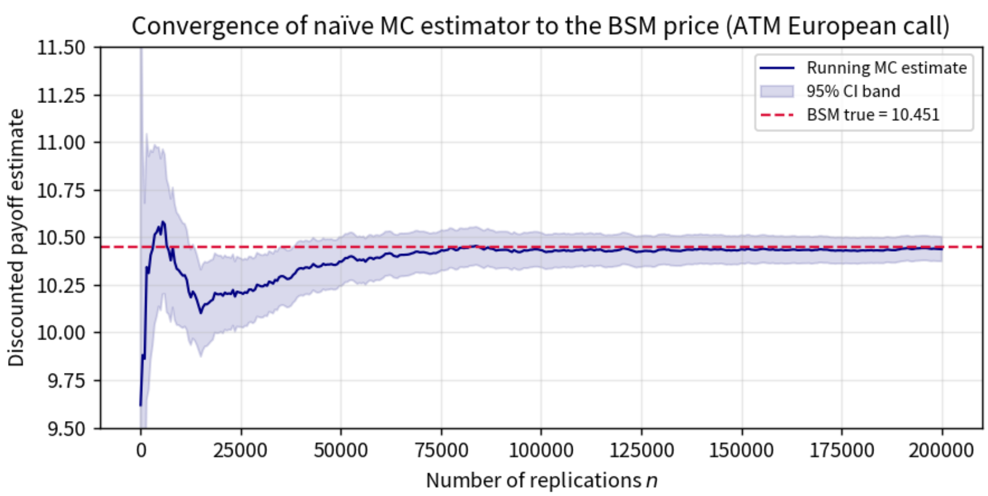

Figure 1. Running naïve-MC estimate (navy) with 95% CI band converging to the BSM price (dashed red) for the ATM European call. The x-axis is the number of replications n.

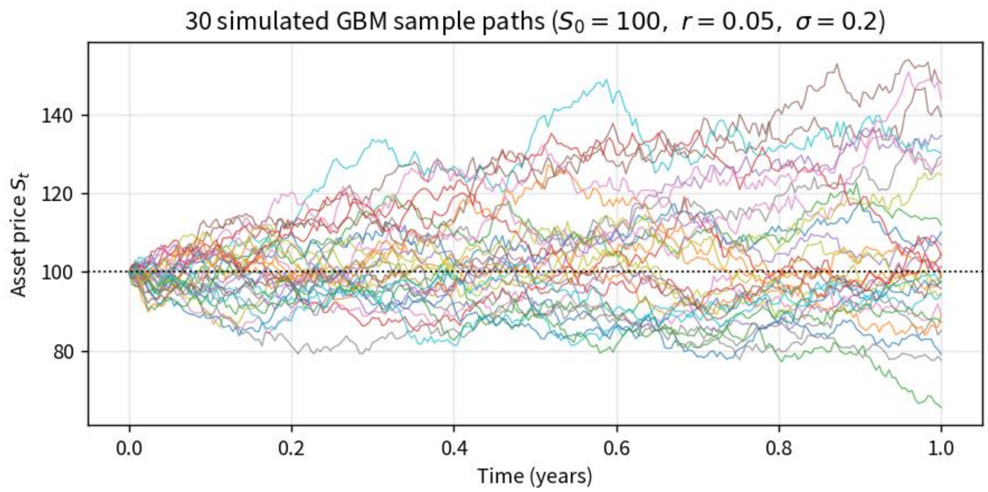

Figure 2. Thirty risk-neutral GBM sample paths (S_0 = 100, r = 0.05, σ = 0.20, T = 1). Discounted terminal payoffs of such paths are averaged to price European options.

A single covering interval is weak evidence of correctness, so we run a formal coverage study: for each of three strikes we generate 1,000 independent MC estimates (n = 5,000 each) and record how often the 95% CI covers the true price. Table 4 reports the average estimate, the average of the per-replication reported standard errors, the empirical (true) standard error across the 1,000 macro-replications, and the coverage rate. The reported SE closely matches the empirical SE and coverage is essentially nominal (0.950–0.953), confirming that our standard-error formula and intervals are correctly calibrated.

K	True price	Avg MC est	Coverage (nominal 0.95)	Avg reported SE	Empirical SE
90.0000	16.6994	16.7155	0.9500	0.2461	0.2441
100.0000	10.4506	10.4421	0.9520	0.2081	0.2109
110.0000	6.0401	6.0378	0.9530	0.1644	0.1645

Table 4. CI-coverage validation: 1,000 macro-replications of n = 5,000 each. Nominal coverage is 0.95; reported and empirical standard errors agree closely.

4.2 Variance-reduction study

Naïve MC error decays only as $n^{-\frac{1}{2}}$, so halving the CI width costs 4× the work. Variance reduction buys precision without more paths. We study two techniques. Antithetic variates pair each shock Z with its negative −Z, inducing negative correlation between paired payoffs. Control variates subtract a mean-zero adjustment built from a correlated quantity with known expectation: for the European call we use the discounted terminal price

($E_Q[e^{-rT}S_T\,] \;=\; S_0$); for the arithmetic-Asian call we use the geometric-average Asian payoff, whose exact price is known from Kemna–Vorst (1990).

Table 5 compares the four estimators for a European call (K = 90) at a fixed budget of $n \;=\; 10^5$. Antithetic variates give a negligible gain here because the call payoff is nearly monotone in Z (little symmetric cancellation to exploit), but the discounted-price control variate cuts variance by ~17×, and combining both is marginally better still. All four estimates agree with the BSM value of 16.699.

Method	**Estimate**	**SE**	**Var. reduction (×)**	**CI low**	**CI high**
Naïve	16.7484	0.0552	1.0000	16.6403	16.8565
Antithetic	16.7137	0.0550	1.0043	16.6059	16.8216
Control variate	16.7137	0.0132	17.3297	16.6877	16.7396
Antithetic + Control	16.7094	0.0132	17.3969	16.6835	16.7354

Table 5. Variance reduction for European call ($K \;=\; 90, n \;=\; 10^5$). 'Var. reduction' is the ratio of naïve to method variance (SE²).

The payoff for path-dependent options is where control variates shine. Table 6 prices the arithmetic-Asian call (K = 100, m = 50) three ways. The geometric-average control variate is almost perfectly correlated with the arithmetic payoff, driving the standard error from 0.0256 down to 0.0007—a variance reduction of roughly 1,300×, equivalent to running about 1.3 billion naïve paths. This is the single most impactful technique in the library and the reason our Asian prices are trustworthy at modest path counts.

Method	**Estimate**	**SE**	**Var. reduction (×)**
Naïve	5.8204	0.0256	1.0000
Antithetic	5.8502	0.0256	0.9989
Geometric control	5.7522	0.0007	1313.2514

Table 6. Variance reduction for the arithmetic-Asian call ($K \;=\; 100, m \;=\; 50, n \;=\; 10^5$). The geometric-average control variate yields ~1,300× variance reduction.

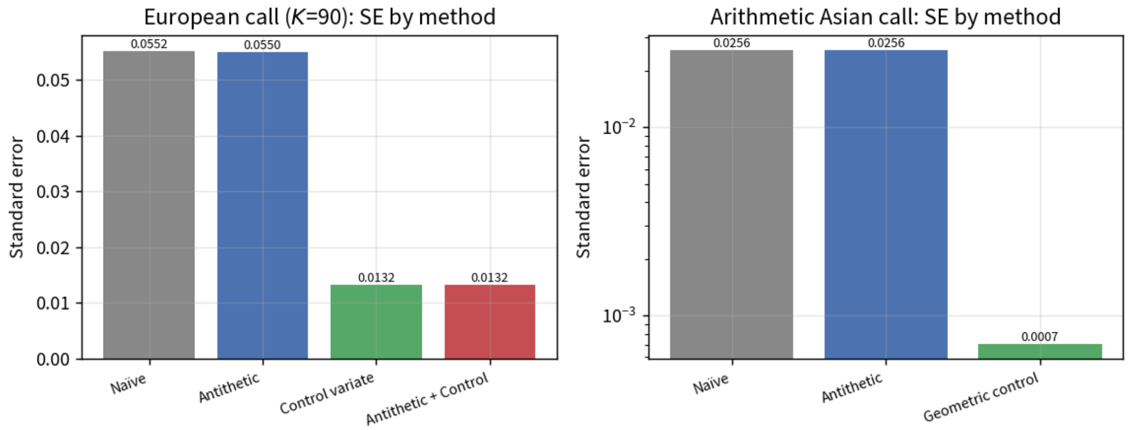

Figure 3. Standard error by method. Left: European call (K = 90), linear scale. Right: arithmetic-Asian call, log scale (note the ~30× SE drop from the geometric control variate).

4.3 American options: least-squares Monte Carlo vs. binomial benchmark

American options may be exercised early, so pricing requires solving an optimal-stopping problem. We implement two methods. The Cox–Ross–Rubinstein (CRR) binomial lattice with 2,000–3,000 steps gives an essentially exact reference by backward induction, taking the maximum of exercise and continuation value at each node. The Longstaff–Schwartz LSM (2001) is the simulation approach: moving backward through time, it regresses realized discounted continuation values on a quadratic basis $\{1, S, S^2\}$ of the in-the-money paths, then exercises wherever the immediate payoff exceeds the fitted continuation value.

Table 7 prices American puts across five strikes. LSM ($n = 2 \times 10^5, m = 50$, antithetic) lands within about 0.7% of the binomial reference at every strike and recovers a strictly positive early-exercise premium (the gap over the European put), which grows with moneyness exactly as theory predicts. LSM prices sit slightly below the lattice, consistent with LSM being a low-biased estimator: an imperfect regressed exercise policy can only be suboptimal, so it underprices the option—a known and desirable property when a conservative lower bound is wanted.

K	European put (BSM)	American (binomial)	American (LSM)	LSM SE	Early-ex premium	|LSM-bin|
90.0000	2.3101	2.4726	2.4486	0.0105	0.1625	0.0240
95.0000	3.7133	4.0136	3.9959	0.0135	0.3003	0.0177
100.0000	5.5735	6.0900	6.0463	0.0160	0.5165	0.0437
105.0000	7.9004	8.7408	8.6990	0.0181	0.8404	0.0419

110.0000	10.6753	11.9732	11.9007	0.0192	1.2979	0.0725

Table 7. American put pricing: LSM vs. binomial reference across strikes (T = 1, σ = 0.20). 'Early-ex premium' = binomial American − European. |LSM − bin| is the absolute pricing gap.

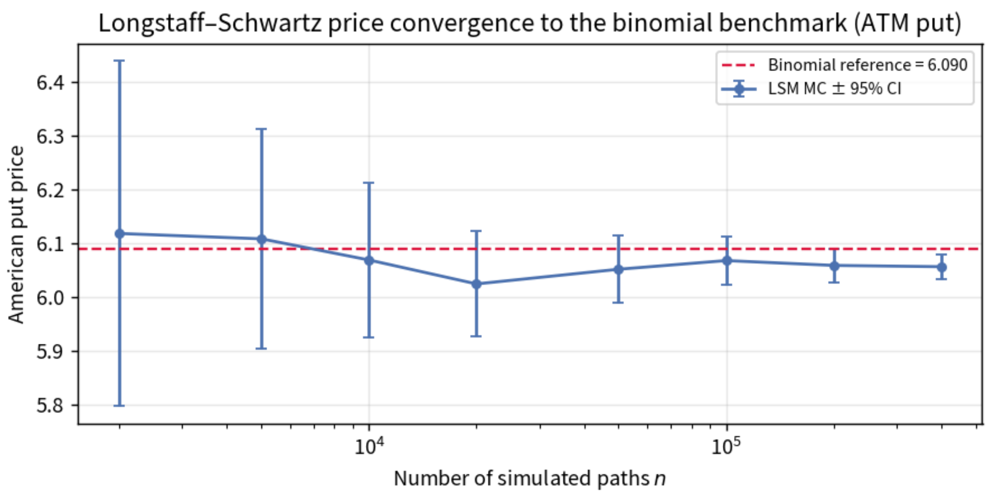

Figure 4. Longstaff–Schwartz price of the ATM American put converging to the binomial benchmark (dashed) as the path count grows; error bars are 95% CIs. The small residual gap reflects LSM's known low bias.

4.4 Greeks: analytical vs. finite-difference Monte Carlo

Risk management requires sensitivities of price to the inputs—the Greeks. Delta and Gamma measure first- and second-order sensitivity to spot; Vega, Theta, and Rho measure sensitivity to volatility, time, and rate. For European options we have exact formulas; for options without closed forms we estimate Greeks by central finite differences of the MC price. The key implementation detail is that the bumped prices must share the same random shocks Z (common random numbers), otherwise the differencing noise swamps the signal (quantified in Section 4.5).

Table 8 compares the analytical Greeks of the ATM European call to finite-difference MC estimates using CRN with n = 10^6. Every estimate matches the analytical value to at least three significant figures; the largest relative error, in Gamma (a second difference, hence the noisiest), is under 2.5%. This validates the finite-difference machinery that the library later applies to exotic payoffs where no analytical Greek exists.

Greek	**Analytical**	**FD MC (CRN)**	**Difference**
Delta	0.63683	0.63674	-0.00009

Gamma	0.01876	0.01921	0.00044
Vega	37.52403	37.53892	0.01488
Theta	-6.41403	-6.41802	-0.00399
Rho	53.23248	53.22316	-0.00932

Table 8. ATM European call Greeks: analytical vs. finite-difference MC with common random numbers ($n = 10^6$). Central differences with bumps hS = 0.5, hσ = 0.005, hr = 0.001, hT = 1/365.

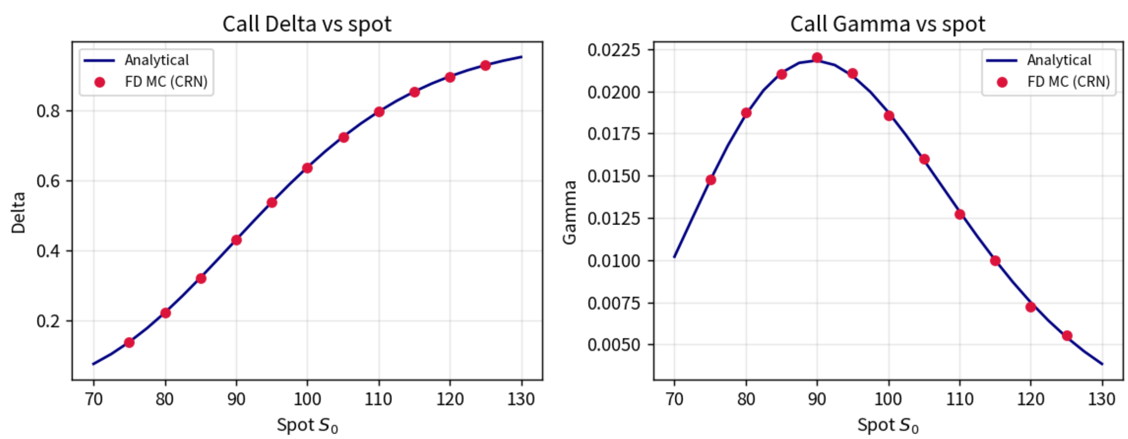

Figure 5. Delta (left) and Gamma (right) versus spot for the European call: analytical curve (navy) with finite-difference MC estimates (red markers) overlaid. The MC points track the analytical profiles across the full moneyness range.

4.5 The value of common random numbers

Section 4.4 relied on CRN; here we quantify why it is essential. We estimate the ATM call Delta by central finite difference two ways—bumping spot up and down with independent random streams, versus reusing the same stream—and repeat each 200 times ($n = 2 \times 10^4$ per price). Table 9 and Figure 6 tell the story: with independent streams the estimator standard deviation is 0.138, so a single estimate is nearly useless; with CRN it collapses to 0.004, a variance reduction of about 1,100×. Because the two bumped prices are almost identical functions of the shared shocks, their difference isolates the sensitivity signal and cancels the common noise. CRN costs nothing extra and is the reason finite-difference Greeks are practical at all.

Estimator	**Mean Delta**	**Std of estimator**	**Var. reduction**
Independent random numbers	0.6338	0.1379	1×
Common random numbers (CRN)	0.6369	0.0042	1104×

Table 9. Finite-difference Delta over 200 macro-replications (true Δ = 0.6368). CRN reduces estimator variance by ~1,100×.

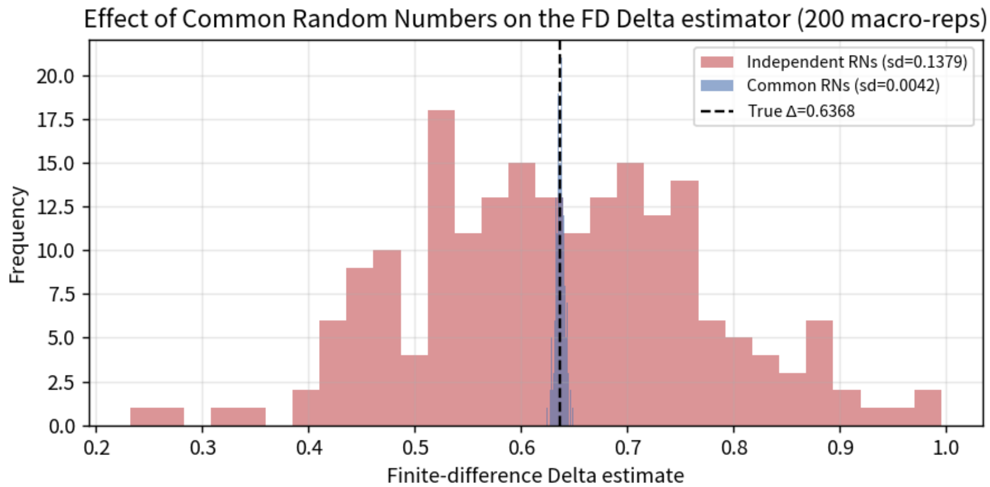

Figure 6. Distribution of the finite-difference Delta estimator over 200 macro-replications. The CRN estimator (blue) is tightly concentrated on the true Delta (dashed), while the independent-stream estimator (red) is badly dispersed.

5 Conclusions and Future Work

5.1 What we found and learned

We built and validated OptionLib, a modular Python library that prices European, Asian, and American options under the BSM framework and reports the statistical quality of every estimate. Four concrete findings emerged. (1) Our naïve MC engine is unbiased: it matches BSM across ten contracts, and a formal coverage study confirms its 95% intervals are correctly calibrated (empirical coverage 0.950–0.953, with reported and empirical standard errors in agreement). (2) Variance reduction is decisive for path-dependent payoffs: a geometric-average control variate cut Asian-option variance by ~1,300×, turning an imprecise estimator into a sharp one at the same cost. (3) Least-squares Monte Carlo prices American puts within ~0.7% of a fine binomial lattice and correctly recovers a positive, moneyness-increasing early-exercise premium, while exhibiting the expected slight low bias. (4) Common random numbers are indispensable for finite-difference Greeks, reducing Delta-estimator variance by ~1,100× at zero extra cost; with CRN our MC Greeks match the analytical values to three significant figures.

The broader lesson is methodological. The same Monte Carlo estimate can be nearly worthless or highly precise depending on whether the analyst exploits structure—antithetic symmetry, a correlated control with known mean, or shared randomness across compared scenarios. Benchmarking every estimator against an analytical or lattice ground truth, and reporting coverage rather than a single interval, is what let us make trustworthy correctness claims rather than merely plausible ones.

5.2 Future work

Several concrete extensions build directly on this codebase. First, replace constant volatility with a stochastic-volatility (Heston) or local-volatility model, which requires simulating a variance process and reintroduces genuine time-discretization bias worth quantifying. Second, add quasi-Monte Carlo (Sobol') sequences and compare their convergence to the pseudo-random $O(n^{-\frac{1}{2}})$ rate observed here. Third, extend LSM to higher-dimensional American basket options and study the sensitivity of the price to the choice and number of regression basis functions, and construct the dual (upper-bound) estimator to bracket the true American price between LSM's lower bound and a martingale upper bound. Fourth, apply the finite-difference CRN Greeks to Asian and American options, where no closed form exists, and validate them by bumping the binomial lattice. Fifth, investigate the one non-covering interval in Table 3 more systematically by studying CI coverage in the deep-OTM regime, where payoff skewness slows the CLT and may warrant bootstrap or importance-sampling intervals. Each of these is a well-scoped problem that our modular architecture supports with minimal new plumbing.

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Appendix A Code Provenance and Reproducibility

All model derivations, code, tables, and figures in this report are the team's own work. The BSM, Kemna–Vorst, CRR-binomial, and Longstaff–Schwartz formulas are standard results from the cited literature; we implemented them from those references rather than copying any third-party code. The simulation engine uses only NumPy (PCG64 RNG, vectorized array operations), SciPy (the normal CDF/PDF), pandas (results tables), and Matplotlib (figures). No external option-pricing library (e.g., QuantLib) was used. The accompanying zip contains the seven modules of Table 2, a README describing each file, and reproduce.py, which regenerates every numerical result and figure here from fixed seeds. Reported seeds appear in reproduce.py; the seed-reuse discipline (shared streams for CRN/control-variate comparisons, independent streams for the coverage study) is documented in Section 3.1.

Appendix B AI-Usage Disclosure

In accordance with the course academic-integrity policy, we disclose all generative-AI use. A large-language-model assistant (OpenAI ChatGPT / Anthropic Claude class) was used as a drafting and editing aid: to help organize the report structure against the rubric, to refine prose for clarity and concision, and to help debug and vectorize portions of the Python implementation. All mathematical content, modeling choices, experimental designs, parameter selections, and interpretations of results were made and verified by the team. Every numerical result reported was produced by our own code executed by the team, and every figure and table was generated from those runs; no results were fabricated or taken from an AI. The team reviewed and takes full responsibility for the entire contents of this report.